Project: TraceMind — Intelligent Multimodal Document Investigation Assistant

Team: HyperNova

Sub-track: 1C - Searching the Way a Human Does

GitHub Repository Link - [TraceMind GitHub Repository]

Demonstration Video Link – [YouTube – TraceMind Demonstration]

Problem Statement

Enterprise knowledge is often fragmented across heterogeneous sources such as PDFs, DOCX files, Markdown, TXT files, scanned documents, images, tables, and diagrams. Traditional keyword search and basic Retrieval-Augmented Generation (RAG) approaches can retrieve semantically similar information, but complex questions often require evidence to be discovered, connected, compared, and validated across multiple documents.

A single retrieval pass may miss critical evidence, fail to distinguish similarly named entities, overlook contradictions between sources, or produce a confident answer even when the available evidence is insufficient.

TraceMind addresses this problem by treating document question answering as an iterative investigation process. Instead of simply retrieving documents and generating an answer, the system plans an investigation, retrieves relevant evidence, analyzes visual information, evaluates evidence quality, detects conflicts, determines whether sufficient evidence has been collected, and performs targeted follow-up retrieval when necessary.

Chosen Sub-track — 1C: Searching the Way a Human Does

TraceMind follows this sub-track by reproducing the way a human researcher investigates a difficult question:

Question → Plan → Search → Examine Evidence → Compare Sources → Identify Conflicts → Identify Missing Evidence → Search Again → Conclude

Solution Overview

TraceMind is a **multimodal, multi-agent document investigation system** designed to produce evidence-grounded answers from large and heterogeneous document collections.

Core Capabilities

Iterative Evidence Retrieval
Rather than relying on a single vector search, TraceMind can retrieve evidence, evaluate what was found, identify missing information, and perform targeted follow-up searches.

Multimodal Investigation
The system can reason over textual documents as well as images, scanned pages, diagrams, and other visual evidence using a dedicated vision model.

Cross-Document Reasoning
Evidence from different documents can be connected through names, aliases, dates, identifiers, document references, and related events.

Evidence Calibration
Retrieved information is analyzed to distinguish directly supported facts from weaker inference or uncertain evidence.

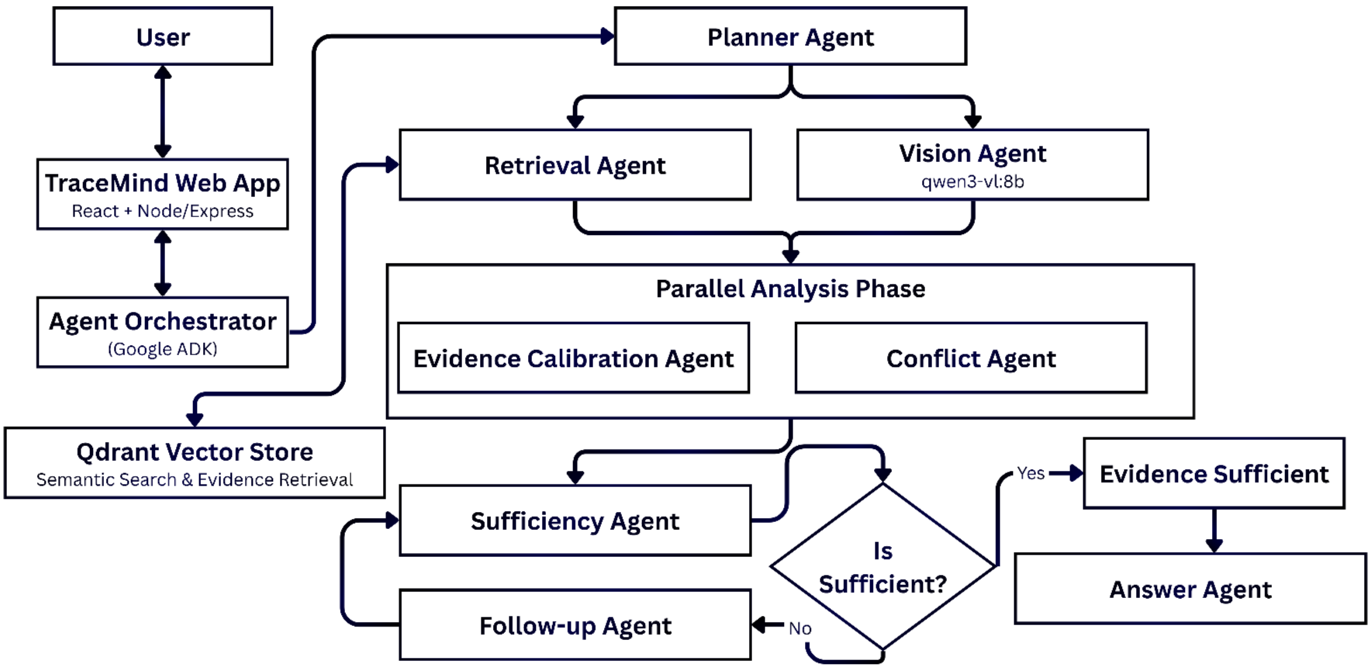
Conflict Detection
Contradictory claims are identified and analyzed. Importantly, similarly named entities are not automatically treated as the same entity without supporting evidence.

Evidence Sufficiency
The system evaluates whether the available evidence is sufficient to answer the question. If not, another investigation round can be initiated.

Grounded Answer Generation
The final answer presents verified findings, uncertainty/conflicting evidence, and source citations rather than producing an unsupported response.

Multi Agent System Architecture & Technical Decisions

Multi Agent System Architecture



Key Technical Decisions

Decision	Why We Choose It
Qdrant Vector Store	Fast semantic retrieval over large document collections and metadata-filtered evidence
Nomic Embeddings	Converts document chunks and queries into semantic vector representations
Google ADK	Provides structured orchestration for the multi-agent investigation workflow
Qwen3 14B	Main reasoning model for planning, evidence analysis, conflict analysis and answer generation
Qwen3-VL 8B	Enables multimodal investigation of visual document evidence
Cloudflare R2	Scalable object storage for uploaded document collections
RunPod + Ollama	Provides controllable GPU-hosted inference infrastructure
LangSmith	Provides tracing, observability and evaluation of the complete agent workflow

What Works, What Doesn't & Limitations

What Works

Multi-source evidence reasoning

TraceMind successfully connects evidence distributed across different documents rather than depending on a single retrieved passage.

Entity-aware conflict analysis

The system can distinguish genuine contradictions from records referring to different entities with similar names.

For example, during evaluation it correctly avoided treating Gloamreach and Gloammarch as automatically referring to the same location.

Multimodal evidence

The Vision Agent allows visual evidence to participate in an investigation when important information exists in images or scanned material.

Evidence sufficiency

The system can recognize when retrieved evidence is insufficient instead of forcing a definitive answer.

Traceable agent execution

LangSmith provides visibility into the investigation pipeline, including planning, query embedding, Qdrant retrieval, evidence analysis, conflict analysis and final answer generation.

What Doesn't

Unnecessary Agent Execution

Running every agent sequentially can increase latency even when some stages are unnecessary.

Improvement: Agent execution was optimized, independent analysis can be performed in parallel where appropriate, and sufficiency checks prevent unnecessary additional retrieval rounds.

Current Limitations

Investigation Latency
Multi-agent reasoning requires several model calls and can therefore take longer than conventional single-pass RAG.

Retrieval Dependency
The quality of an investigation still depends on whether relevant evidence can be successfully retrieved from the indexed corpus.

AI Usage Disclosure

The generative AI-assisted development tool Antigravity was used during the development of TraceMind for brainstorming, code assistance, debugging, prompt refinement, UI development, architecture exploration, and documentation support. The team designed the final system architecture, selected and integrated the technologies, implemented and tested the application, evaluated system behavior, and made the final engineering decisions.

The TraceMind application itself uses Qwen-family language and vision models as core components of its agentic investigation pipeline. These models are orchestrated through Google ADK and evaluated and observed using LangSmith.

Team Members Contribution

Team Member	Role	Key Contributions
Jazeel Jaufer	UI/UX Designer	User research, user flows, interface design and design system.
Sathalogithasiva Hariram	Frontend Developer	React/Vite development, reusable components, application interfaces and API integration.
Ahamed Umar Jiffry	Backend Developer	Node/Express APIs, authentication, document processing, MongoDB and cloud storage integration.
Jegatheesan Risikesan	Agentic AI & Infrastructure Engineer	Multi-agent architecture, Google ADK, RAG/Qdrant, Qwen models, RunPod/Ollama and LangSmith evaluation.